

HANBURY RIVER, NT, Canada

WMO: 719630

Lat: 63.600N

Lon: 105.130W

Elev: 1041

StdP: 14.15

Time Zone: -7.00 (NAM)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-42.8	-39.8	-50.2	0.3	-42.5	-47.0	0.4	-39.9	35.9	-8.5	31.6	-15.7	5.8	270	0.970

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	74.9	59.4	71.2	57.0	67.4	55.6	61.5	71.0	59.1	67.7	57.0	65.0	12.3	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.9	74.5	64.0	55.5	68.0	62.3	53.2	62.5	60.1	28.0	70.8	26.3	67.7	24.9	65.1	78.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 35.5	28.5	24.5	-45.9	80.9	3.5	3.1	-48.5	83.2	-50.5	85.0	-52.5	86.7	-55.1	88.9		
(5)			-47.0	63.8	2.9	6.3	-49.1	68.4	-50.8	72.1	-52.5	75.6	-54.6	80.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	15.2	-20.4	-18.9	-10.7	3.8	24.5	47.0	56.1	52.0	39.3	21.1	-0.7	-13.2	(6)
(7)		DBStd	29.51	12.79	12.53	13.52	12.58	11.40	8.42	6.40	6.76	8.38	9.92	12.26	12.71	(7)
(8)		HDD50	13105	2183	1928	1881	1385	790	156	16	56	332	896	1522	1960	(8)
(9)		HDD65	18190	2648	2348	2346	1835	1255	542	283	405	770	1361	1972	2425	(9)
(10)		CDD50	401	0	0	0	0	0	65	206	118	12	0	0	0	(10)
(11)		CDD65	10	0	0	0	0	0	1	7	2	0	0	0	0	(11)
(12)		CDH74	132	0	0	0	0	0	22	80	30	0	0	0	0	(12)
(13)		CDH80	9	0	0	0	0	0	2	6	1	0	0	0	0	(13)
(14)	Wind	WSAvg	10.7	10.4	10.1	10.0	9.8	10.3	10.3	11.2	11.0	12.7	11.7	11.2	10.1	(14)
(15)	Precipitation	PrecAvg	11.5	0.4	0.4	0.5	0.4	0.7	1.2	1.8	2.2	1.7	1.1	0.6	0.4	(15)
(16)		PrecMax	16.3	0.7	0.7	1.0	1.1	2.1	3.3	4.2	3.7	2.8	1.9	1.3	1.0	(16)
(17)		PrecMin	8.3	0.1	0.1	0.1	0.1	0.1	0.2	0.8	0.9	0.5	0.5	0.3	0.2	(17)
(18)		PrecStd	1.6	0.2	0.2	0.2	0.2	0.4	0.6	0.7	0.7	0.6	0.3	0.3	0.2	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	18.1	17.7	28.7	36.2	56.5	77.2	80.3	77.9	64.8	46.3	30.6	21.3	(19)
(20)			MCWB	17.7	17.4	27.4	33.8	45.4	61.2	62.0	59.8	54.3	43.5	30.2	20.8	(20)
(21)		2%	DB	11.7	11.5	20.2	32.9	48.2	71.7	76.1	72.5	59.9	40.3	25.9	17.0	(21)
(22)			MCWB	11.3	10.9	19.3	31.7	42.0	55.5	60.4	58.5	51.7	38.8	25.6	16.5	(22)
(23)		5%	DB	5.0	6.1	15.1	28.1	42.3	67.1	72.6	68.0	55.7	36.3	21.6	11.8	(23)
(24)			MCWB	4.4	5.5	14.4	26.7	38.5	54.9	57.6	56.1	49.1	35.0	21.1	11.3	(24)
(25)		10%	DB	-1.3	1.0	10.1	23.7	38.3	62.9	69.3	63.8	51.4	33.3	17.7	6.3	(25)
(26)			MCWB	-1.9	0.4	9.3	22.6	35.5	51.8	56.4	54.3	47.1	32.6	17.2	5.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	17.0	23.1	34.0	46.7	66.8	64.9	63.0	56.5	44.4	29.9	21.0	(27)
(28)			MCD B	17.3	17.2	23.8	35.5	53.6	68.7	74.5	71.5	62.3	47.2	30.2	21.4	(28)
(29)		2%	WB	10.1	9.3	17.8	31.3	41.9	59.4	62.0	60.1	52.7	39.1	24.5	16.8	(29)
(30)			MCD B	10.4	9.9	18.6	32.5	47.3	67.7	72.7	68.6	58.5	40.3	24.8	17.3	(30)
(31)		5%	WB	3.8	3.6	13.3	26.4	38.8	55.7	59.8	57.7	49.8	35.2	19.7	11.0	(31)
(32)			MCD B	4.3	4.1	14.0	27.3	42.2	65.7	69.1	65.6	54.0	36.1	20.1	11.5	(32)
(33)		10%	WB	-2.4	-1.3	8.3	22.3	35.6	52.8	57.9	55.7	47.2	32.8	15.4	5.4	(33)
(34)			MCD B	-2.0	-0.7	9.1	23.3	37.9	61.8	66.1	62.3	50.9	33.4	15.8	5.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.6	14.3	15.9	17.9	14.2	18.5	17.6	15.8	12.5	10.2	13.2	14.1	(35)
(36)			MCDBR	17.8	19.1	18.1	18.7	18.2	26.6	23.0	22.8	19.8	11.9	14.9	19.5	(36)
(37)			MCWBR	17.8	20.0	17.7	17.0	12.2	14.9	10.5	10.9	12.3	10.5	16.8	20.2	(37)
(38)		5% WB	MCDBR	18.1	19.6	18.0	17.7	17.4	25.1	20.8	19.9	17.9	11.7	16.3	19.9	(38)
(39)			MCWBR	17.9	19.4	17.6	16.6	12.1	16.7	11.6	11.0	12.3	10.4	16.2	19.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.161	0.234	0.247	0.292	0.309	0.336	0.353	0.328	0.296	0.255	0.173	0.143	(40)	
(41)		taud	1.990	2.169	2.268	2.175	2.293	2.445	2.417	2.468	2.544	2.333	1.964	1.933	(41)	
(42)		Ebn at Noon	161	228	276	281	287	280	271	269	259	218	141	91	(42)	
(43)		Edh at Noon	9	19	26	36	35	31	31	27	21	16	9	4	(43)	
(44)	All-Sky Solar Radiation	RadAvg	60	299	860	1569	1938	1840	1668	1227	716	231	79	25	(44)	
(45)		RadStd	4	14	58	96	111	132	108	91	90	53	12	2	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page